Aryan Saraf

(408) 590-5570 | aryansaraf05@gmail.com | LinkedIn | Portfolio

EDUCATION

University of Southern California – M.S. Computer Science Jan 2025 – Dec 2026

Coursework: Networked Systems for Cloud Computing, Analysis of Algorithms, Applied NLP, Multimedia Systems Design

University of Mumbai – B.E. Information Technology Aug 2019 – May 2023

SKILLS

Programming: Go, Python, Java, C#, Bash, SQL

Frameworks: .NET, Unity, PyTorch

Networking and Systems: TCP/IP, BGP, OSPF, ECN, DCTCP, SDN (OpenFlow), DNS (resolution, record types, TTL caching, Cloud DNS), Linux tc, netlink, Mininet, Open vSwitch

Cloud and Infrastructure: GCP, Apigee API Gateway, Kubernetes, Docker, Terraform, Kafka, GitHub Actions, ELK

Databases: PostgreSQL, MySQL, Cassandra, ScyllaDB, MongoDB, Redis, Elasticsearch

EXPERIENCE

Software Engineering Intern | Easley-Dunn Productions, Inc. – Los Angeles, CA May 2025 – Aug 2025

- Built a C# .NET backend API integrated with a Unity client to power facility management in a sports simulation application
- Held API latency under 50 ms (local) with real-time synchronization between the backend database and the Unity client
- Reduced integration time ∼40% by delivering reusable Unity prefabs and handler scripts for developer onboarding
- Designed facility upgrade flows with data-driven SQLite storage and JSON configurations

DevOps Engineer | Jio Platforms Limited – Mumbai, India Oct 2023 – Dec 2024

- Architected a real-time change-data-capture (CDC) pipeline streaming NoSQL changes into Elasticsearch for live analytics
- Improved API response times 30% by configuring Apigee API Gateway with OAuth and token auth for high-traffic APIs
- Reduced deployment time ∼80% by automating GCP provisioning with Terraform across 4 standardized environments
- Cut platform latency 30% with Azure DevOps CI/CD pipelines and zero-downtime Kubernetes rollouts
- Saved $7,000+ monthly by decommissioning idle infrastructure; ran vulnerability and malware scans on 100+ servers
- Operated 7 HA database engines (Cassandra, ScyllaDB, MongoDB, Redis) across dev, QA, UAT, pre-prod and production

PROJECTS

Fabric Microburst Telemetry | GitHub | *Go, Python, Mininet, Linux tc, netlink* Aug 2026 – Sep 2026

- Detected 100% of 8 ms congestion bursts with 1 ms queue polling vs. 1% at 1 s, where byte counters reported 17% peak utilization on a link saturated during every burst
- Built a zero-allocation Go agent sampling switch-queue depth over raw netlink every 100 µs, plus a synchronized incast generator emulating AI collective traffic on a leaf-spine fabric
- Found ECN marking cut median peak queue 29% but left median collective completion unchanged and p99 15% worse; diagnosed a 10.9 ms shaper stall caused by GSO super-packets

Latency-Aware Load Balancer | GitHub | *Go, Mininet, OpenFlow* Jan 2026 – May 2026

- Cut p99 tail latency 3.8× (570 ms to 150 ms vs. Least Connections) under straggler injection with a Go load balancer that routes by real-time active probing (Prequal) on a two-level fat-tree
- Sent up to 39% fewer probes at equal p99 with a controller that adapts probe rate to server heterogeneity
- Isolated network delay as the driver of the gain: on a flat topology, all 4 routing policies tied at ∼43 ms p99

Adaptive Query Reformulation (AQR-RAG) | *Python, DistilRoBERTa, Google Trends API* Aug 2025 – Dec 2025

- Flagged weak queries at 82.4% accuracy (0.90 PR-AUC) with fine-tuned DistilRoBERTa to trigger RAG query rewriting
- Injected real-time Google Trends context into ambiguous queries, aligning reformulations with trending terminology